Year 11 Higher

REVISION PACK

Handling Data 4

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Q	Working	Answer
questionnaires		
1.1		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
1.2		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 7; no box for 0**
1.3		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6 KD; no box for 0**
1.4		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
1.5		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 30 KD; no box for 0**
1.6		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 4; no box for 0**
averages from a frequency table		
1.7	**(a)** $\text{highest frequency} = 8$ **(b)** $0 \times 3 + 1 \times 7 + 2 \times 8 + 3 \times 5 + 4 \times 2 = 46$ $46 \div 25 = 1.84$	**(a) 2** **(b) 1.84**
1.8	**(a)** $\text{highest frequency} = 9$ **(b)** $1 \times 2 + 2 \times 9 + 3 \times 6 + 4 \times 4 + 5 \times 4 = 74$ $74 \div 25 = 2.96$	**(a) 2** **(b) 2.96**

1.9	**(a)** highest frequency $= 8$ **(b)** $0 \times 6 + 1 \times 5 + 2 \times 4 + 3 \times 8 + 4 \times 2 = 45$ $45 \div 25 = 1.8$	**(a) 3** **(b) 1.8**
1.10	**(a)** highest frequency $= 9$ **(b)** $0 \times 5 + 1 \times 9 + 2 \times 4 + 3 \times 1 + 4 \times 1 = 24$ $24 \div 20 = 1.2$	**(a) 1** **(b) 1.2**
1.11	**(a)** highest frequency $= 8$ **(b)** $0 \times 2 + 1 \times 4 + 2 \times 8 + 3 \times 4 + 4 \times 2 = 40$ $40 \div 20 = 2$	**(a) 2** **(b) 2**
1.12	**(a)** highest frequency $= 8$ **(b)** $0 \times 3 + 1 \times 8 + 2 \times 6 + 3 \times 2 + 4 \times 1 = 30$ $30 \div 20 = 1.5$	**(a) 1** **(b) 1.5**
a missing number from the mean		
1.13	$5 \times 16 = 80$ $12 + 14 + 19 + 20 = 65$ $80 - 65 = 15$	**15**
1.14	$6 \times 11 = 66$ $8 + 9 + 13 + 14 + 10 = 54$ $66 - 54 = 12$	**12**
1.15	$5 \times 23 = 115$ $18 + 25 + 27 + 20 = 90$ $115 - 90 = 25$	**25**
1.16	$5 \times 12 = 60$ $9 + 11 + 14 + 15 = 49$ $60 - 49 = 11$	**11**
1.17	$4 \times 20 = 80$ $18 + 25 + 16 = 59$ $80 - 59 = 21$	**21**

1.18	$6 \times 15 = 90$ $12 + 18 + 14 + 20 + 11 = 75$ $90 - 75 = 15$	**15**

Handling Data 4 · Pie charts (Lesson 4.3)

Q	Working	Answer
pie chart angles		
2.1	$360 \div 90 = 4$ $25 \times 4 = 100$	**100°**
2.2	$360 \div 120 = 3$ $38 \times 3 = 114$	**114°**
2.3	$360 \div 40 = 9$ $13 \times 9 = 117$	**117°**
2.4	$360 \div 72 = 5$ $20 \times 5 = 100$	**100°**
2.5	$360 \div 30 = 12^\circ$ for each student $84 \div 12 = 7$	**7**
2.6	$360 \div 180 = 2^\circ$ for each person $130 \div 2 = 65$	**65**

Handling Data 4 · Misleading charts (Lesson 4.4)

Q	Working	Answer
what is misleading?		
2.7	Any two from:	**Vertical axis starts at 100, not 0** **The bar for Wed is wider than the others** **No title**
2.8	Any two from:	**The vertical scale is not even (10, 20, 40, 80)** **No label on the vertical axis** **No title**
2.9	Any two from:	**Vertical axis starts at 40, not 0** **No label on either axis** **No title**